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A simplified space vector modulation for the single-phase active power decoupling quasi-Z-source inverter

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ABSTRACT

The double-line frequency ripple power of the single-phase quasi-Z source inverter (qZSI) will result in a large designed qZS impedance on the dc side, which can be greatly reduced by the coupled-type active power decoupling (APD) method. However, the traditional sinusoidal pulse width modulation (SPWM) for the APD-qZSI needs extravagant APD storage capacitance and may cause abnormal operation under low load conditions. Classical space vector modulation (SVM) has higher flexibility and dc-link voltage utilization, but it cannot be directly applied to the APD-qZSI. In this paper, a simplified SVM method for the coupled-type APD-qZSI is proposed, where the switching time sequence is transformed to carrier-based pulse width modulation, so the shoot-through control of the qZSI is easy to implement. Next, the passive component parameters and normal operation range of the traditional SPWM and the proposed SVM are analyzed and compared. The proposed SVM has a smaller APD storage capacitance but a slightly larger ac side filter inductance. In addition, the decreased current of the APD branch under SVM can help suppress the abnormal operation of the inverter, so the APD-qZSI can operate at a lower load condition. Finally, experimental results validate the effectiveness and superiority of the proposed SVM method.

1. Introduction

The quasi-Z source inverter (qZSI) is a promising topology in renewable energy power generation applications such as photovoltaic (PV) and fuel cells [1–3]. Compared with the traditional two-stage voltage source inverter (VSI), the qZSI can boost and buck voltage in a single stage by using the additional shoot-through (ST) state. Bridge legs are allowed to be short-circuited, which improves the system reliability and output ac voltage quality.

An unavoidable challenge for the single-phase qZSI is to mitigate the double-line-frequency ripple power, which is a common issue for all single-phase inverters. This kind of ripple power will degrade system performance [4,5], for example, increasing odd harmonic components of the grid-tie currents, reducing the maximum power point tracking (MPPT) efficiency, and so on. To limit the ripple power within the engineering tolerance range, [6] gives the qZS impedance design methods by analyzing the relationship of the ripple current/voltage versus qZS impedance. These methods are classified as passive decoupling solutions, which will result in large designed inductance and capacitance, and low power density of the system.

A more prospective method for this issue is the active power decoupling (APD) technology. [7–9] have reviewed a variety of APD topologies and control methods of the traditional single-phase VSI. The ripple power can be generally balanced by introducing an additional power compensation branch to the dc-link. As a result, the dc side of the inverter only flows through dc power, the small passive component is enough to handle the residual switching-frequency ripple. This approach offers promising applications as it significantly reduces the overall weight and cost of the inverter without compromising its functionality. The APD methods of single-phase qZSI are first introduced in [10] and have been proven to be effective in reducing the inductance and capacitance of the qZS network. There generally exist few types of APD-qZSI topologies [11], such as buck-type [12], interleaved type [13], boost-type [14], and coupled-type [10]. Among them, the coupled type has attracted more attention because the voltage of the APD capacitor is bipolar and easy to track by the control system. [10] discusses the parameter design (including qZS impedance and APD branch impedance) and control strategies of the coupled-type APD-qZSI, where the influence of the ST duty cycle on system parameter design is fully concerned. To improve the dynamic performance of the coupled-type

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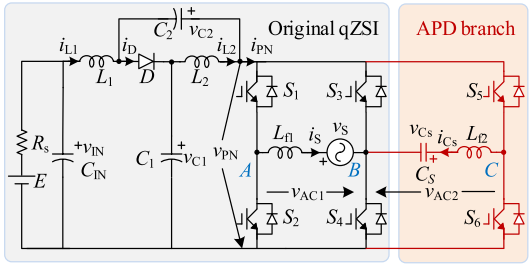


Fig. 1. The topology of the single-phase coupled-type APD-qZSI.

APD-qZSI, linear proportional-resonant (PR) control [15], direct power control [16], and model predictive control [17] are developed. However, little consideration is given to the effect of different modulation methods on the APD-qZSI.

For the traditional single-phase qZSI, the abnormal operation state will occur when the qZS inductance or load is too small [18]. The abnormal operation should be avoided or suppressed since it will lead to undesired dc-link voltage drops, resulting in grid side current distortion and poor system control performance. This problem is still existed in the APD-qZSI and is strongly related to the modulation methods. Under the traditional SPWM, the maximum APD capacitor voltage is lower than the product of the dc-link voltage and modulation index [10]. Thus, extravagant APD storage capacitance is needed [19], and the higher APD branch current can easily lead the APD-qZSI to occur abnormal operation. However, this issue has not been addressed before.

Besides the SPWM, the SVM is another popular modulation method in three-phase converter systems [20]. Due to its higher flexibility and dc-link voltage utilization, we speculate that the SVM may further reduce the APD storage capacitance and suppress the abnormal operation of the APD-qZSI. However, it has not been reported in the literature. One of the reasons is that the coupled-type APD-qZSI is an unbalanced and atypical three-phase system, the traditional SVM cannot be directly applied. Another reason is that additional ST zero vectors need to be inserted into the existing eight vectors [21,22], thus the implementation of the SVM for the APD-qZSI is much more complex.

This paper proposes a simplified SVM method for the coupled-type APD-qZSI, and smaller APD storage capacitance and wider normal operation range are achieved compared with the traditional SPWM. The rest of the paper is organized as follows. Section II introduces the fundamental of the coupled-type APD-qZSI, then proposes a general control strategy and a simplified SVM method. Section III compares the APD storage capacitance and filter inductance under SPWM and SVM. The normal operation range of these two modulation methods is analyzed in Section IV. Experimental results are given in Section V. Finally, Section VI concludes this study.

2. The Operation, Control, and modulation of the coupled-type APD-qZSI

2.1. Operation principle

Fig. 1 shows the topology of the single-phase coupled-type APD-qZSI, where an APD branch consists of a half-bridge leg (denoted as phase leg "*C*"), a filter inductor L_{f2} and a storage capacitor C_S is connected to the original qZSI circuits.

The original qZSI has two operating states: ST state and non-shoot through (NST) states. At the ST state, the H-bridge turns to short-circuit to boost the dc voltage. At the NST state, the H-bridge converter operates normally and transfers energy to the ac grid. In steady state, the dc-link peak voltage V_{PN} is [1]

$$V_{PN} = V_{C1} + V_{C2} = \frac{V_{IN}}{1 - 2D_0} \tag{1}$$

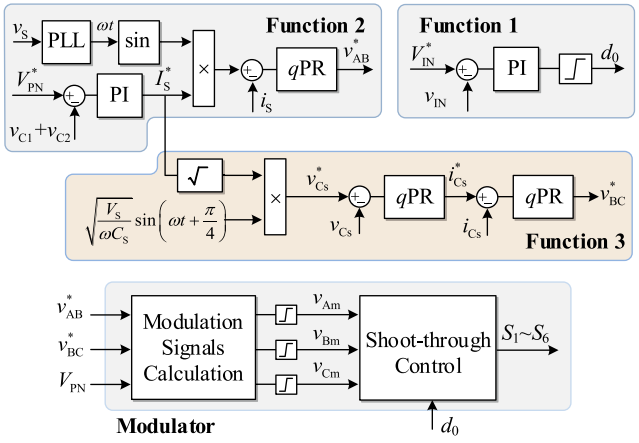


Fig. 2. Overall control scheme for APD-qZSI.

where D_0 is the ST duty cycle; V_{IN} is the input voltage.

Assume the grid voltage v_S and grid side current i_S to be

$$v_S = V_S \sin\omega t \ , \ i_S = I_S \sin\omega t \tag{2}$$

where V_S and I_S are the voltage and current peak values, respectively; ω is the grid angular frequency. Since the voltage drop on the filter inductor is relatively small, ignore the power on L_{f1}, the instantaneous power p_{AB} delivered to the grid is

$$p_{AB} = v_S i_S = \underbrace{0.5 V_S I_S}_{p_{AB_dc}} - \underbrace{0.5 V_S I_S \cos(2\omega t)}_{p_{AB_2\omega}} \tag{3}$$

The p_{AB} consists of two parts: a dc power (p_{AB_dc}) and a second-order harmonic (2ω) ripple power ($p_{AB_2\omega}$). The $p_{AB_2\omega}$ varies at twice the grid frequency and will cause the 2ω voltage and current ripple in qZS impedance. Therefore, large capacitors and inductors are usually necessary.

The APD branch is used to balance the 2ω ripple power. Suppose the storage capacitor voltage v_{Cs} and current i_{Cs} to be

$$\begin{cases} v_{Cs} = V_{Cs} \sin(\omega t + \gamma) \\ i_{Cs} = I_{Cs} \cos(\omega t + \gamma) = \omega C_S V_{Cs} \cos(\omega t + \gamma) \end{cases} \tag{4}$$

where V_{Cs} and I_{Cs} are the capacitor voltage and current peak values, respectively, and $I_{Cs} = \omega C_S V_{Cs}$; γ is the phase angle difference between v_{Cs} and v_S. Ignore the power on L_{f2}, the instantaneous power delivered to the APD branch is

$$p_{CB} = v_{Cs} i_{Cs} = 0.5 \omega C_S V_{Cs}^2 \sin(2\omega t + 2\gamma) \tag{5}$$

To completely balance the 2ω ripple power, p_{CB} needs to be controlled equal to $p_{AB_2\omega}$. Therefore, the required ripple power decoupling conditions are

$$V_{Cs} = \sqrt{\frac{V_S I_S}{\omega C_S}}, \gamma = \frac{1}{4}\pi \tag{6}$$

If (6) satisfied, the 2ω ripple power can be directly transferred to the APD branch, and the inverter dc side only flows through dc power. As a result, only a small qZS capacitance and inductance are enough to suppress the residual high-frequency voltage and current ripple.

2.2. Control strategy

Three control functions are designed to maintain a constant input voltage, guarantee a unity power factor with a constant dc-link voltage, and suppress the dc-link voltage ripple. The overall control scheme is shown in Fig. 2.

Control function 1 is a simple PI feedback controller of the input voltage v_{IN}, its output signal is the ST duty cycle d_0. Control function 2

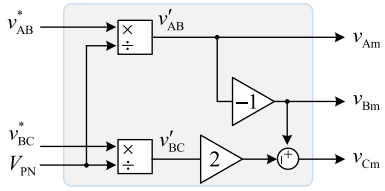


Fig. 3. Modulation signals calculation block of the traditional SPWM.

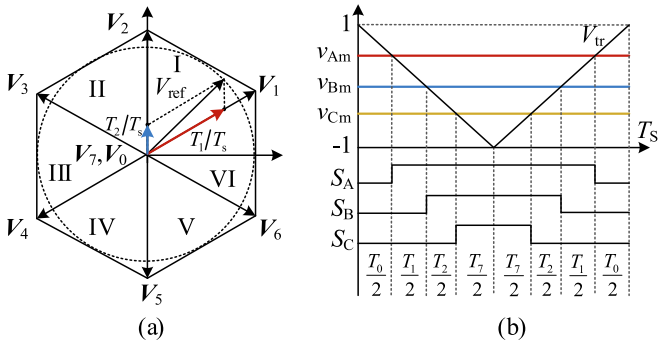


Fig. 4. Line-voltage-based SVM method. (a) Basic voltage space vectors and (b) switching time sequence.

consists of an outer-loop dc-link peak voltage controller and an inner-loop grid side current controller. The feedback signal of the outer-loop controller is generated by summing v_{C1} and v_{C2} according to (1), its output is the amplitude reference of the grid side current, denoted as I^*_S. The inner-loop controller adopts a quasi-proportional-resonant (qPR) controller, and its output signal is the reference line voltage between phase legs *A* and *B*, denoted as v^*_{AB}. Control function 3 consists of two qPR controllers in series. The reference of the outer loop is

$$v^*_{Cs} = \sqrt{\frac{V_S I^*_S}{\omega C_S}} \sin(\omega t + \frac{1}{4}\pi) \tag{7}$$

where ωt is generated by the phase-locked-loop (PLL) of the grid voltage, and I^*_S is from Function 2. The output of the inner loop is the reference line voltage between phase legs *B* and *C*, denoted as v^*_{BC}.

Next, the line voltage reference v^*_{AB} and v^*_{BC} are sent to the "Modulation Signals Calculation" block to obtain the three-phase modulation signals (v_{Am}, v_{Bm}, and v_{Cm}). Then, the ST duty cycle d_0 and the modulation signals are sent to the "Shoot-through Control" block to generate the final switching pulse of $S_1 \sim S_6$.

The ST state can be realized by a variety of techniques, such as Simple Boost Control, Constant Boost Control, and Maximum Boost Control [23]. However, the shoot-through duty cycle of the latter two methods includes low-frequency components, which can result in additional current and voltage ripples on the DC side. Hence, these methods are not suitable for single-phase qZSI. Since the Simple Boost Control is simplest and most widely used [10], it is applied in the "Shoot-through Control" block.

The implementation of the "Modulation Signals Calculation" block is discussed below.

Table 1
Relationship between *M* and *N*.

M	1	2	3	4	5	6
Sector *N*	VI	II	I	IV	V	III

2.3. Modulation signals calculation block: traditional SPWM

Various SPWM techniques can be applied to qZSI, such as hybrid SPWM [24] and sinusoidal pulse width amplitude modulation (SPWAM) [25], some of which can additionally reduce the qZS impedance size (unlike the APD approach). In order to simplify the analysis, this paper adopts the most typical SPWM method shown in Fig. 3 and compares it as a reference group. The line-voltage references are firstly divided by the nominal dc-link voltage V_{PN} to generate the per-unit values v'_{AB} and v'_{BC}, respectively. Normally, v'_{AB} and v'_{BC} vary from − 1 to 1. According to the SPWM techniques, the three-phase modulation signals are $v_{Am} = -v_{Bm} = v'_{AB}$ and $v_{Cm} = v_{Bm} - 2v'_{BC}$.

2.4. Modulation signals Calculation Block: The proposed simplified SVM

The coupled-type APD-qZSI can be analyzed as an unbalanced three-phase inverter and the output three-phase line voltages meet $v_{AB} + v_{BC} + v_{CA} = 0$. The $\alpha\beta$ transformation is

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} v_{AB} \\ v_{BC} \\ v_{CA} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \sqrt{2} \end{bmatrix} \begin{bmatrix} v_{AB} \\ v_{BC} \end{bmatrix} \tag{8}$$

The line-voltage-based SVM method has eight voltage space vectors (from V_0 to V_7) and six sectors (marked as I, II, III… VI), as shown in Fig. 4(a). The voltage vector length is $\sqrt{2}\ V_{PN}$.

Assume the reference voltage V_{ref} sits in the sector I, by using the transformation equation (8), the time intervals of active vectors V_1 and V_2 can be expressed as

$$\begin{cases} T_1 = \dfrac{\sqrt{6}\, v_\alpha T_S}{3 v_{PN}} = \dfrac{T_S v_{AB}}{v_{PN}} = T_S v'_{AB} \\ T_2 = \dfrac{\left(v_\beta - \sqrt{3}\left/3 v_\alpha\right.\right) T_S}{\sqrt{2}\, v_{PN}} = \dfrac{T_S v_{BC}}{v_{PN}} = T_S v'_{BC} \end{cases} \tag{9}$$

where T_S is the switching period; v_α and v_β are the projections of reference voltage on α and β axes, respectively. It can be seen from (9), the time intervals are functions of per-unit values of the line voltage.

Fig. 4(b) shows the switching time sequence in sector I when using the seven-segment SVM, where T_0 and T_7 are the time intervals of zero vectors V_0 and V_7, respectively, and $T_0 = T_7 = (T_S - T_1 - T_2)/2$; S_A, S_B, and S_C are the upper switch gate signals of the three-phase legs; V_{tr} is the triangular carrier wave, whose amplitude is ± 1 and period is T_S; v_{Am}, v_{Bm}, and v_{Cm} are the modulation signals. S_A, S_B, and S_C can be generated by comparing v_{Am}, v_{Bm}, and v_{Cm} with V_{tr}, From Fig. 4(b), the relationships between modulation signals and time intervals are

$$\begin{cases} v_{Am} = 1 - \dfrac{2T_0}{T_S} = \dfrac{T_1 + T_2}{T_S} = v'_{AB} + v'_{BC} \\ v_{Bm} = 1 - \dfrac{2(T_0 + T_1)}{T_S} = \dfrac{T_2 - T_1}{T_S} = v'_{BC} - v'_{AB} \\ v_{Cm} = 1 - \dfrac{2(T_0 + T_1 + T_2)}{T_S} = \dfrac{-T_1 - T_2}{T_S} = -\left(v'_{AB} + v'_{BC}\right) \end{cases} \tag{10}$$

Thus, the SVM switching time sequence can be easily transformed into carrier-based SPWM. Applying the same rules to other sectors, the

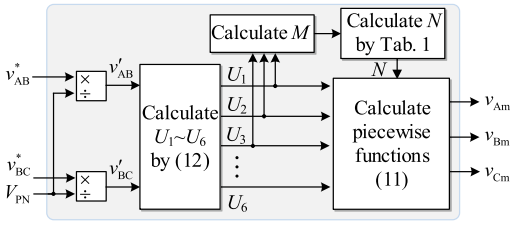


Fig. 5. Modulation signals calculation block of the simplified SVM.

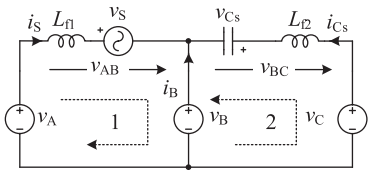


Fig. 6. Averaged model of the ac side APD-qZSI in NST state.

three-phase modulation signals can be calculated as

$$v_{Am}=\begin{cases} U_3\ ,\ N\ =\ \text{I or IV} \\ U_5\ ,\ N\ =\ \text{II or V} \\ U_1\ ,\ N\ =\ \text{III or VI} \end{cases}$$
$$v_{Bm}=\begin{cases} -U_4\ ,\ N\ =\ \text{I or IV} \\ U_2\ ,\ N\ =\ \text{II or V} \\ -U_1\ ,\ N\ =\ \text{III or VI} \end{cases} \quad (11)$$
$$v_{Cm}=\begin{cases} -U_3\ ,\ N\ =\ \text{I or IV} \\ -U_2\ ,\ N\ =\ \text{II or V} \\ -U_6\ ,\ N\ =\ \text{III or VI} \end{cases}$$

where $U_1 \sim U_6$ are temporary variables, defined as

$$\begin{aligned} &U_1=v'_{AB}, U_2=v'_{BC}, U_3=v'_{AB}+v'_{BC} \\ &U_4=v'_{AB}-v'_{BC}, U_5=2v'_{AB}+v'_{BC}, U_6=v'_{AB}+2v'_{BC} \end{aligned} \quad (12)$$

This paper uses $U_1 \sim U_3$ to judge the sector, the principle is directly given here: if $U_1 > 0$, then $A = 1$, otherwise $A = 0$; if $U_2 > 0$, then $B = 1$, otherwise $B = 0$; if $U_3 > 0$, then $C = 0$, otherwise $C = 1$, define $M = 4C + 2B + A$, the relationship between M and sector N is shown in Table 1.

Fig. 5 shows the "Modulation Signals Calculation" block diagram using the proposed simplified SVM methods. Firstly, this block calculates the per-unit value of the input line-voltage reference. Next, the variables $U_1 \sim U_6$ are calculated using (12). Then, sector N is judged by Table 1 using $U_1 \sim U_3$. Finally, the three-phase modulation signals are generated by (11) using $U_1 \sim U_6$ and N.

From Fig. 5, we can find: (1) The realization of the proposed SVM is very simple because only a few algebraic functions are contained; (2) The ST control methods of the traditional SPWM can be also suitable for the proposed SVM since the modulation signals of the SVM are obtained.

3. Passive component Parameters comparison of the SPWM and SVM method

3.1. APD storage capacitance

According to (6), to balance the 2ω ripple power, the larger the storage capacitor voltage V_{Cs}, the smaller the required storage capacitance C_S. Therefore, calculating the maximum value of V_{Cs} under different modulation methods is the key to the selection of C_S. Ignoring the high-frequency switching components, the averaged equivalent circuit of the ac side of the APD-qZSI in the NST state is shown in Fig. 6. v_A, v_B, and v_C represent the equivalent phase voltages.

Neglect the voltage drop on filter inductors L_{f1} and L_{f2}, the voltage equations of circuit loops "1" and "2" are

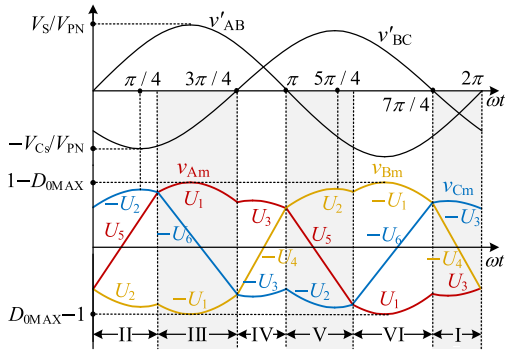


Fig. 7. Modulation signals of the simplified SVM.

$$\begin{cases} v_{AB}=L_{f1}\dfrac{di_S}{dt}+v_S\approx V_S\sin\omega t \\ v_{BC}=-L_{f2}\dfrac{di_{Cs}}{dt}-v_{Cs}\approx -V_{Cs}\sin\left(\omega t+\dfrac{\pi}{4}\right) \end{cases} \quad (13)$$

which indicates that if the APD-qZSI is well controlled, the line voltage references v^*_{AB} and v^*_{BC} sent to the "Modulation Signals Calculation" block are approximately equal to v_S and $-v_{Cs}$, respectively.

(1) Storage capacitance requirement of the SPWM.

From the traditional SPWM method in Fig. 3, the three-phase modulation signals can be calculated as

$$\begin{cases} v_{Am}=-v_{Bm}=M_A\sin\omega t \\ v_{Cm}=M_C\sin(\omega t+\varphi_C) \end{cases} \quad (14)$$

with

$$M_A=\frac{V_S}{V_{PN}}, M_C=\frac{\sqrt{\left(\sqrt{2}\,V_{Cs}-V_S\right)^2+2V_{Cs}^2}}{V_{PN}} \quad (15)$$

$$\tan\varphi_C=\frac{\sqrt{2}V_{Cs}}{\sqrt{2}V_{Cs}-V_S} \quad (16)$$

where M_A and M_C denote the modulation index of phases A and C, respectively; φ_C is the phase angle difference between v_{Am} and v_{Cm}.

In the Simple Boost ST control method, the modulation index needs to meet

$$M_A \leqslant 1-D_{0MAX} \quad , M_C \leqslant 1-D_{0MAX} \quad (17)$$

where D_{0MAX} is the maximum value of the ST duty cycle. To keep (17) always holds, in the design of a closed-loop controller, d_0 is limited to [0, D_{0MAX}], and v_{Am}, v_{Bm}, and v_{Cm} are limited to [D_{0MAX} − 1, 1 − D_{0MAX}]. From (17) and (15), we can obtain

$$\begin{cases} V_S \leqslant V_{PN}(1-D_{0MAX}) \\ V_{Cs} \leqslant \dfrac{1}{2}\sqrt{(1-D_{0MAX})^2V_{PN}^2-\dfrac{1}{2}V_S^2}+\dfrac{\sqrt{2}}{4}V_S \end{cases} \quad (18)$$

To maximize the utilization of the dc-link voltage, set $V_S = V_{PN}(1 - D_{0MAX})$. From (18) and (6), the maximum voltage amplitude of the storage capacitor under the traditional SPWM method, the minimum storage capacitance required for eliminating the 2ω ripple power, and the mini minimum current amplitude of the storage capacitor are

$$V_{Cs,Max}=\frac{\sqrt{2}}{2}V_S, C_{S,Min}=\frac{2I_S}{\omega V_S}, I_{Cs,Min}=\sqrt{2}I_S \quad (19)$$

(2) Storage capacitance requirement of the SVM.

According to the SVM method shown in Fig. 5, the three-phase modulation signals are piecewise functions of the input line-voltage reference v^*_{AB} and v^*_{BC} defined in (13). The modulation signals under SVM are shown in Fig. 7.

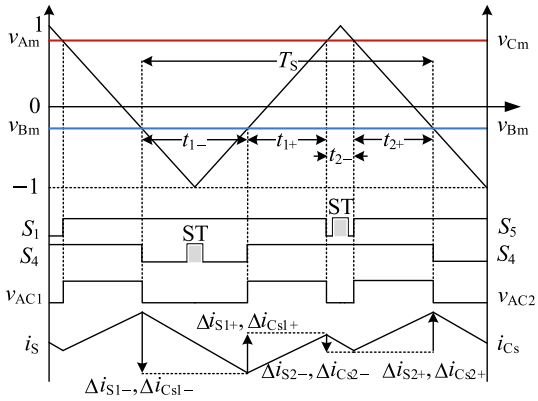


Fig. 8. Waveform of the filter inductor current ripple.

It can be seen that the modulation signals are quasi-saddle waves with two wave heads. The boundary phase angles of sector I/II, sector III/IV, sector IV/V, sector VI/I, and sector I/II are 0, 3π/4, π, 7π/4, and 2π, respectively. The boundary phase angles of sector II/III and sector V/VI are

$$\theta_{II/III} = -\arctan\frac{\sqrt{2}V_{Cs}}{\sqrt{2}V_{Cs}-2V_S}, \theta_{V/VI} = \pi + \theta_{II/III} \tag{20}$$

It can be easily proved that when $\pi/4 < \theta_{II/III} < \pi/2$ and $5\pi/4 < \theta_{V/VI} < 3\pi/2$ (that is when $V_S \geq V_{Cs} > \sqrt{2}V_S/2$), the v_{Cm} has the peak values at $\omega t = \pi/4$ and $\omega t = 5\pi/4$, and the absolute peak value is V_{Cs}/V_{PN} (the amplitude of $-U_2$). The v_{Am} has the peak values at $\omega t = \pi/2$ and $\omega t = 3\pi/2$. The absolute peak value is V_S/V_{PN} (the amplitude of U_1). From Fig. 7 and (17), we can obtain

$$V_{Cs} \leqslant V_{PN}(1 - D_{0MAX}) \qquad , V_S \leqslant V_{PN}(1 - D_{0MAX}) \tag{21}$$

Set $V_S = V_{PN}(1 - D_{0MAX})$ and $V_{Cs} = V_{PN}(1 - D_{0MAX})$, then the maximum voltage amplitude of the storage capacitor, the minimum storage capacitance, and the minimum current amplitude of the storage capacitor under the SVM method are

$$V_{Cs,Max} = V_S, \qquad C_{S,Min} = \frac{I_S}{\omega V_S}, \qquad I_{Cs,Min} = I_S \tag{22}$$

Comparing (22) and (19), the storage capacitor under SVM is only half of that under the traditional SPWM.

3.2. AC side filter inductance

The filter inductors L_{f1} and L_{f2} in APD-qZSI are used to suppress the high-frequency ripple of the grid-side current and the storage capacitor current, respectively. Generally, the filter inductor value is chosen to attain a desired current ripple. Different modulation methods (SPWM and SVM) will lead to different filter inductances.

The instantaneous output voltages of the *A-B* leg and *C-B* leg, denote as V_{AC1} and V_{AC2}, are pulse waves whose amplitude changes from V_{PN} to 0 (positive period) or $-V_{PN}$ to 0 (negative period) alternately. The ripple characteristics of these two periods are similar, this paper only analyzes the positive one.

If the inverter is in the NST state, there have $v_{AC1} = v_{AC2} = V_{PN}$, i_S and i_{Cs} increase linearly since v_S and v_{Cs} change very small within a carrier period. If the switches are in zero states or the inverter is in the ST state, there have $v_{AC1} = v_{AC2} = 0$. i_S and i_{Cs} decrease linearly. The inductor current ripples are

$$\Delta i_{S+} = \frac{v_{PN} - v_S}{L_{f1}}\Delta t_+, \Delta i_{S-} = \frac{-v_S}{L_{f1}}\Delta t_- \tag{23}$$

$$\Delta i_{Cs+} = \frac{v_{PN} - v_{Cs}}{L_{f2}}\Delta t_+, \Delta i_{Cs-} = \frac{-v_{Cs}}{L_{f2}}\Delta t_- \tag{24}$$

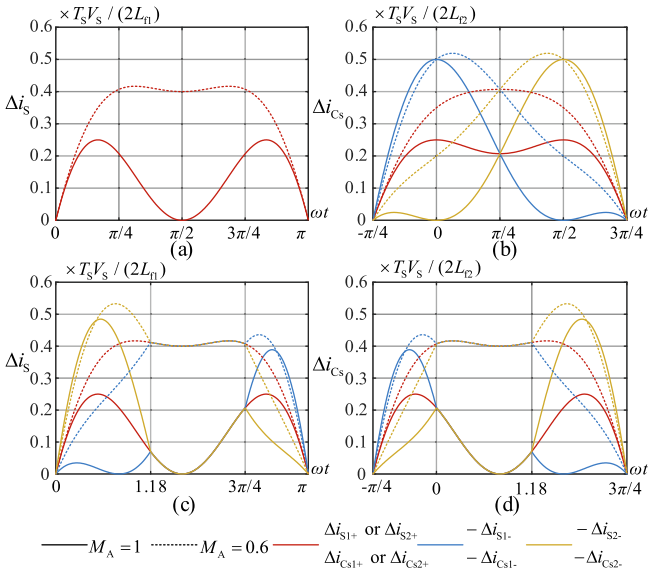


Fig. 9. Current ripples of (a) i_S under SPWM, (a) i_{Cs} under SPWM, (a) i_S under SVM, and (d) i_{Cs} under SVM.

where Δi_{S+}, Δi_{S-} and Δi_{Cs+}, Δi_{Cs-} are the positive/negative ripples of i_S and i_{Cs}, respectively. Δt_+ and Δt_- are the subintervals of the current increasing and decreasing process, respectively.

Fig. 8 shows the diagram of the filter inductor current ripple based on the modulation technique. The inductor current increases twice and decreases twice within one switching period. The subintervals can be expressed as

$$\begin{cases} \Delta t_{1+} = \Delta t_{2+} = \dfrac{T_S}{4}(v_{Am} - v_{Bm}) \\ \Delta t_{1-} = \dfrac{T_S}{2}(1 + v_{Bm}) \ , \ \Delta t_{2-} = \dfrac{T_S}{2}(1 - v_{Am}) \end{cases} \tag{25}$$

$$\begin{cases} \Delta t_{1+} = \Delta t_{2+} = \dfrac{T_S}{4}(v_{Cm} - v_{Bm}) \\ \Delta t_{1-} = \dfrac{T_S}{2}(1 + v_{Bm}) \ , \ \Delta t_{2-} = \dfrac{T_S}{2}(1 - v_{Cm}) \end{cases} \tag{26}$$

where subscripts 1 and 2 denote the first and second current increasing/decreasing processes.

According to the modulation signals generated from the "Modulation Signals Calculation" block in Fig. 3 and Fig. 5, the expressions of the current ripples under different modulation methods are given in the Appendix. The waveforms of these ripples are shown in Fig. 9.

The maximum value of Δi_S under SPWM is $T_S V_S/(8L_{f1}M_A)$. Thus, the selection of L_{f1} is

$$L_{f1} \geqslant \frac{V_{PN}T_S}{8\lambda_1 I_S} \tag{27}$$

where λ_1 is the current ripple coefficient of i_S.

It can be seen from Fig. 9(b)~(d) that the maximum values of these current ripples are nearly equal to 0.5 when M_A varies from 0.6 to 1. Thus, the selection of L_{f2} under SPWM is

$$L_{f2} \geqslant \frac{V_S T_S}{4\lambda_2 I_{Cs}} = \frac{V_S T_S}{4\sqrt{2}\lambda_2 I_S} \tag{28}$$

where λ_2 is the current ripple coefficient of i_{Cs}. Since i_{Cs} do not require a small total harmonic distortion (THD), to save cost, λ_2 can be larger than λ_1.

For the SVM method, the filter inductances are

$$L_{f1} \geqslant \frac{V_S T_S}{4\lambda_1 I_S}, L_{f2} \geqslant \frac{V_S T_S}{4\lambda_2 I_{Cs}} = \frac{V_S T_S}{4\lambda_2 I_S} \tag{29}$$

Table 2
Nominal circuit parameters.

Parameters	Value	Parameters	Value
Grid peak voltage, V_S	311 V	Grid peak Current, I_S	19.3A
Grid frequency, f	50 Hz	Dc-link peak voltage, V_{PN}	500 V
Input voltage, V_{IN}	200 V ~ 240 V	Input current, I_{L1}	15A
Carrier frequency	20 kHz	ST duty cycle, D_0	0.3

From (27) to (29), the minimum filter inductor L_{f1} under SVM is $2M_A$ times of that under SPWM, and the minimum filter inductor L_{f2} under SVM is $\sqrt{2}$ times of that under SPWM.

4. Normal operation range

For the traditional single-phase qZSI, the abnormal operation state will occur when the current of the qZS diode is discontinuous. Too small qZS inductance or load can cause this current discontinuity. Since the coupled-type APD-qZSI has added a third phase leg and an *LC* branch, its abnormal operation characteristics will be different from that of the traditional single-phase qZSI and are closely related to the modulation methods.

The qZS diode current at the NST state is [1,18]

$$i_D = 2i_{L1} - i_{PN} \tag{30}$$

where i_{L1} and i_D are the currents of inductor L_1 and diode D, respectively; i_{PN} is the input current of the three-phase bridge legs. In the NST state, the bridge legs have eight switching states, from (0,0,0) to (1,1,1). The value of i_{PN} changes in $\pm\, i_S$, $\pm\, i_{Cs}$, $\pm(i_S + i_{Cs})$, and 0 according to different switching sequences.

Ignore the high-frequency ripple of i_{L1} and the ac side current, to avoid the abnormal operation, i_D in (30) needs to be greater than 0, which is

$$\frac{V_S I_S^{'}}{V_{IN}} - i_{PN} > 0 \Rightarrow \frac{V_S I_S^{'}}{V_{IN}} > Max\left(\left|i_S^{'}\right|, \left|i_{Cs}^{'}\right|, \left|i_S^{'} + i_{Cs}^{'}\right|\right) \tag{31}$$

where the superscripts represent the non-rated current values that are less than or equal to their rated ones. From (4) and (6), the storage capacitor current amplitude $I_{Cs}^{'}$ under different grid side current amplitude $I_S^{'}$ is

$$I_{Cs}^{'} = \sqrt{\omega C_S V_S I_S^{'}} \tag{32}$$

For the traditional SPWM method, if the storage capacitance under rated power is taken as (19), then $I_{Cs}^{'}$ is

$$I_{Cs}^{'} = \sqrt{2 I_S^{'} I_S} \tag{33}$$

It can be proved that the maximum value of the three-phase current amplitude is $I_{Cs}^{'}$. Substituting (33) into (31), we can obtain

$$\frac{I_S^{'}}{I_S} > \frac{2V_{IN}^2}{V_S^2} \tag{34}$$

Define the left-side of (34) as load rate. To operate the APD-qZSI at a lower load without an abnormal operation state, input voltage V_{IN} needs to be designed much less than V_S, which will lead to a high shoot-through duty cycle and low dc voltage utilization.

For the proposed SVM method, taking the storage capacitance under rated power as (22), we can obtain

$$I_{Cs}^{'} = \sqrt{I_S^{'} I_S} \tag{35}$$

To avoid abnormal operation, the load rate needs to meet

$$\frac{I_S^{'}}{I_S} > \frac{V_{IN}^2}{V_S^2} \tag{36}$$

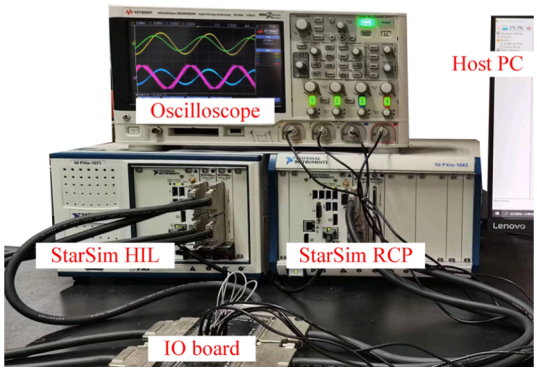
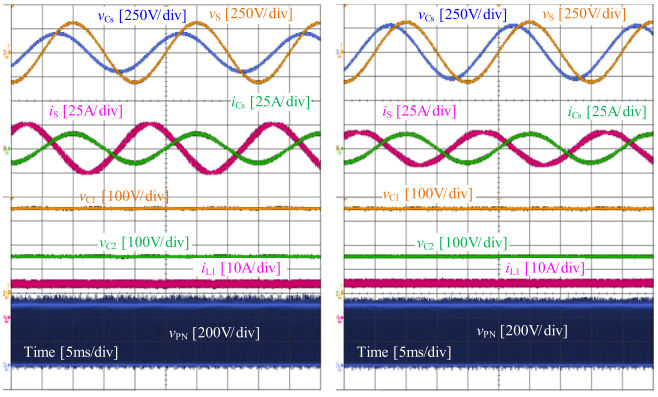


Fig. 10. Experimental platform of the APD-qZSI.



(a) Typical SPWM (b) Proposed SVM

Fig. 11. Steady-state experimental results of the APD-qZSI under (a) typical SPWM and (b) the proposed SVM.

It can be seen from (34) and (36), the load rate range of the SVM is twice that of the SPWM, indicating that the APD-qZSI under SVM can normally operate at a lower load rate.

5. Experimental verification

5.1. Parameter design

The nominal circuit parameter of the APD-qZSI is listed in Table 2. According to the qZS impedance design methods of the coupled-type APD-qZSI [10], the values for L_1 and L_2 are 1.2mH, and the values for C_1 and C_2 are 100μF. These values are chosen to achieve a desired current ripple ratio of 20 % and voltage ripple ratio of 2 %. For the SPWM method, if the storage capacitor is designed according to (19), the capacitance C_S is 395 μF. The rated voltage and current amplitude of the storage capacitor are $V_{Cs} = 220$ V and $I_{Cs} = 27.3$ A, respectively. When $\lambda_1 = 10$ % and $\lambda_2 = 30$ %, the filtering inductance values are $L_{f1} = 1.6$ mH and $L_{f2} = 0.5$ mH, respectively. On the other hand, for the SVM method, if the storage capacitor is designed according to (22), the capacitance value is $C_S = 197.5$ μF. The rated voltage and current amplitude of the storage capacitor are $V_{Cs} = 311$ V and $I_{Cs} = 19.3$ A, respectively. The filtering inductance values are $L_{f1} = 2$ mH and $L_{f2} = 0.7$ mH. Compared to the typical SPWM, the proposed SVM method requires a slightly larger AC-side filtering inductance but much smaller storage capacitors. Consequently, it is anticipated that the volume and cost of APD-qZSI under SVM control will be reduced.

5.2. Experimental tests

The validity of the proposed method is demonstrated by a real-time

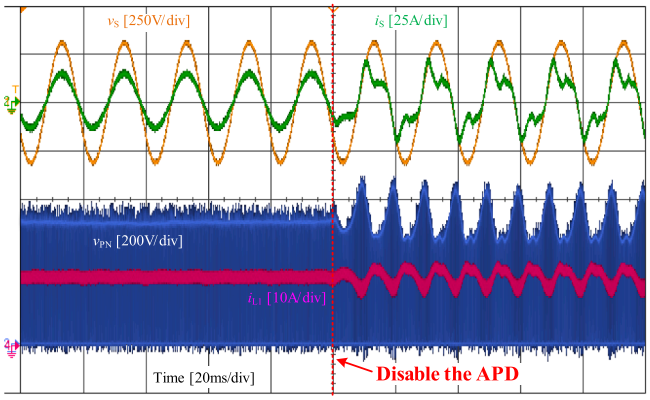


Fig. 12. Experimental results of the APD-qZSI under SPWM when suddenly breaking the APD branch.

experimental test system shown in Fig. 10. The main circuits of the APD-qZSI are built on StarSim HIL (hardware-in-the-loop) platform, and the controllers are implemented on StarSim RCP (rapid control prototyping) platform.

The steady-state experimental results of the coupled-type APD-qZSI are shown in Fig. 11. The voltages and currents of the storage capacitors are well controlled as sinusoidal waves. Low distorted grid currents with unity power factor are achieved. The maximum current ripples of the i_S or i_{Cs} under SPWM and SVM are approximately equal, indicating the correctness of the filter inductor selection methods. It can be also found that no obvious 2ω fluctuation can be seen from qZS inductor currents i_{L1} and dc-link voltages v_{PN}, since the low-frequency ripple power has been diverted to the storage capacitors of the APD branches.

Fig. 12 shows the experimental results when suddenly blocking the APD branch. This experimental test focuses solely on the SPWM method. This is because the modulation signals of the phase *A* and *B* in the SVM method are intertwined with the modulation signals of the phase *C*. Breaking the APD branch can result in the failure of the SVM. The purpose of this experimental test is to compare the performance with and without the APD branch, in order to emphasize the effectiveness of the APD method. It can be seen from Fig. 12 that after breaking the APD branch, the low-frequency ripple of the qZS inductor current is increased from 2.9A to 8.3A, and the low-frequency ripple of the dc-link peak voltage is increased from 3.7 V to 243 V. Due to the sharp 2ω fluctuation of the dc-link voltage, the grid side current is distorted and the measured THD is changed from 2.26 % to 43.68 %. These experimental results validate the effectiveness of APD-qZSI in suppressing the 2ω fluctuations of dc-side voltage and current.

Fig. 13 shows the experimental results when the input voltage reference changes from 200 V to 220 V, causing the load rate changes from 1 to 0.37. According to (34) and (36), after the input voltage changed, the required minimum load rate for the SPWM and SVM are 0.72 and 0.36, respectively, indicating the abnormal operation state will occur under the SPWM. It can be seen from Fig. 13(a), after input voltage changes, i_{L1} occurs some undesired low-frequency ripples when using SPWM. These ripples are different from the traditional 2ω ripples. They are caused by dc-link voltage drops during abnormal operation state, as shown in the enlarged view of v_{PN}, which will distort the grid side current and deform the APD effects. However, for the SVM method, preferable APD performance is always maintained, as shown in Fig. 13 (b), since no abnormal operation state exists.

6. Conclusion

This paper proposes a simplified space vector modulation method for the coupled-type APD-qZSI applied in PV generation systems, comparative analysis is carried out for the SPWM and SVM methods in terms of APD storage capacitance, ac side filter inductance, and normal operation range. The following conclusions can be drawn:

(1) The proposed SVM method transfers the switching time sequence to modulation signals which only contain a few algebraic functions. Thus, the implementation of the SVM for the APD-qZSI is simple;
(2) The APD method under the proposed SVM method is effective for suppressing the 2ω ripple power and reducing the qZS inductance and capacitance, achieving a higher power density of the inverter;
(3) The proposed SVM method enhances the voltage utilization of the APD branch, enabling a higher voltage and lower current in the APD branch. As a result, it allows for a smaller storage capacitance and a wider normal operation range compared to the traditional SPWM method.

CRediT authorship contribution statement

Rui Zhang: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Shunliang Wang:** Conceptualization, Visualization, Resources, Supervision, Writing – review & editing. **Yuan Li:** Investigation, Software, Validation, Data curation, Writing – original draft. **Junpeng Ma:** Conceptualization, Visualization, Resources, Supervision, Writing – review & editing. **Peng Wang:** Software, Validation, Writing – review & editing. **Tianqi Liu:** Resources, Supervision, Writing – review & editing.

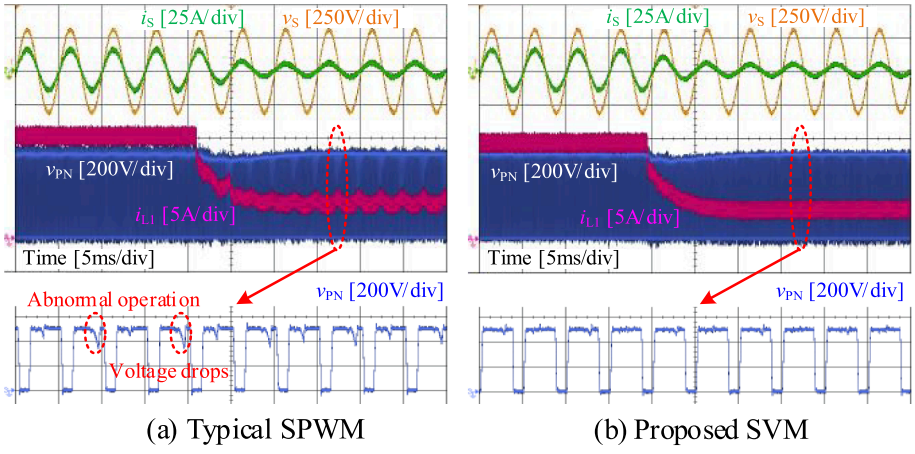


Fig. 13. Transient-state experimental results when the input voltage reference is changed (the output power becomes small).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A

Define the temporary variable a and b as

$$a = \sin\omega t \quad , \quad b = \sin\left(\omega t + \frac{\pi}{4}\right) \tag{37}$$

For the SPWM methods, the expression of the current ripples of i_S are

$$\begin{aligned}\Delta i_{S1+} = \Delta i_{S2+} = -\Delta i_{S1-} = -\Delta i_{S2-} \\ = \frac{T_S V_S a}{2L_{f1}}(1 - M_A a)\end{aligned} \tag{38}$$

where $\omega t \in [0, \pi]$. Assuming the storage capacitor is designed as (19), the expression of the current ripples of i_{Cs} are

$$\begin{cases}\Delta i_{Cs1+} = \Delta i_{Cs2+} = \dfrac{T_S V_S b}{4L_{f2}}\left(\sqrt{2} - M_A b\right) \\ \Delta i_{Cs1-} = -\dfrac{\sqrt{2}\,T_S V_S b}{4L_{f2}}(1 - M_A a) \\ \Delta i_{Cs2-} = -\dfrac{\sqrt{2}\,T_S V_S b}{4L_{f2}}\left(1 + M_A a - \sqrt{2} M_A b\right)\end{cases} \tag{39}$$

where $\omega t \in [-\pi/4, 3\pi/4]$.

For the SVM methods, Assuming the storage capacitor is designed as (22), the expression of the current ripples of i_S are

$$\begin{aligned}\Delta i_{S1+} = \Delta i_{S2+} = \frac{T_S V_S a}{2L_{f1}}(1 - M_A a) \\ \Delta i_{S1-} = -\frac{T_S V_S a}{2L_{f1}} \times \begin{cases} 1 - M_A b & , \quad \omega t \in [0, \theta_{II/III}] \\ 1 - M_A a & , \quad \omega t \in [\theta_{II/III}, 3\pi/4] \\ 1 - M_A (a + b) & , \quad \omega t \in [3\pi/4, \pi] \end{cases} \\ \Delta i_{S2-} = -\frac{T_S V_S a}{2L_{f1}} \times \begin{cases} 1 - M_A (2a - b) & , \quad \omega t \in [0, \theta_{II/III}] \\ 1 - M_A a & , \quad \omega t \in [\theta_{II/III}, 3\pi/4] \\ 1 - M_A (a - b) & , \quad \omega t \in [3\pi/4, \pi] \end{cases}\end{aligned} \tag{40}$$

where $\omega t \in [0, \pi]$ and $\theta_{II/III} = 1.18$ according to (20). The expression of the current ripples of i_{Cs} are

$$\begin{aligned}\Delta i_{Cs1+} = \Delta i_{Cs2+} = \frac{T_S V_S b}{2L_{f2}}(1 - M_A b) \\ \Delta i_{Cs1-} = -\frac{T_S V_S b}{2L_{f2}} \times \begin{cases} 1 - M_A (a + b) & , \quad \omega t \in [-\pi/4, 0] \\ 1 - M_A b & , \quad \omega t \in [0, \theta_{II/III}] \\ 1 - M_A a & , \quad \omega t \in [\theta_{II/III}, 3\pi/4] \end{cases} \\ \Delta i_{Cs2-} = -\frac{T_S V_S b}{2L_{f2}} \times \begin{cases} 1 + M_A (a - b) & , \quad \omega t \in [-\pi/4, 0] \\ 1 - M_A b & , \quad \omega t \in [0, \theta_{II/III}] \\ 1 + M_A (a - 2b), & \quad \omega t \in [\theta_{II/III}, 3\pi/4] \end{cases}\end{aligned} \tag{41}$$

where $\omega t \in [-\pi/4, 3\pi/4]$.

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